



CS & IT ENGINEERING

Discrete Mathematics

✓
Set Theory and Algebra

DPP- 01 Discussion

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#Q. Consider the following statements. How many statements are true? = 4

- | | | | | | |
|---|------|---------------------------|---|---|-----------------------------------|
| { | I. | $\phi \in \phi$ | F | } | $\emptyset \notin \emptyset$ |
| | II. | $\phi \subset \phi$ | F | | $\emptyset \not\subset \emptyset$ |
| | III. | $\phi \subseteq \phi$ | T | | $\emptyset \subseteq \emptyset$ ✓ |
| | IV. | $\phi \in \{\phi\}$ | T | | $\emptyset \in \{\emptyset\}$ |
| | V. | $\phi \subset \{\phi\}$ | T | | |
| | Vi. | $\phi \subseteq \{\phi\}$ | T | | |

4 - True
2 - False

Subsets
are

✓
 $\{\}$ ✓
 $\{\emptyset\}$

$\{1, 2\}$

$\{\}, \{1\}, \{2\}$

$\{1, 2\}$

[NAT]



#Q. If a set A has 63 proper subsets, then what is the cardinality of A?

= 6

Ans

if $|A| = n$

then no. of proper subsets = $2^n - 1$

$$2^n - 1 = 63$$

$$2^n = 64 = 2^6$$

$$\boxed{n = 6}$$

[MCQ]

Subsets of odd cardinality = 64 = $2^{n-1} \Rightarrow 2^6 = 2^{n-1}$
of set A
 $\therefore 6 = n-1 \Rightarrow n=7$
 $\therefore |A|=7$



#Q. If a set A has 64 subsets of odd cardinality, then what is |A|?

- A 6
- B 63
- C 7
- D 128

if $|A|=n, n \geq 1$,
then
Subsets of odd cardinality = 2^{n-1}
Subsets of even cardinality = 2^{n-1}

if $|A|=0$,
i.e. $A = \{\}$

$\Rightarrow$ then
No. of subsets with odd cardinality = 0
No. of subsets with even cardinality = 1

$A = \{a\}$

$|A|=1$

subsets = $2^1 = 2$

$\{\}$ = 1 subset of even cardinality

$\{a\}$ = 1 subset of odd cardinality

$A = \{a, b\}$

$|A|=2$

Subsets = $2^2 = 4$

$\{a\}, \{b\}$
odd

$\{\}, \{a, b\}$
even

[NAT]



#Q.

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

Subsets with
no even integer =

$$2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 = 2^6$$

How many subset of $\{1, 2, 3, \dots, 11\}$ contain at-least one even integer?

= 1984

Ans

No. of subsets
with at least
one even
integer =

Total no.
of
subsets

Subsets with
no
even integer

$$= 2^{|\text{Set}|} - 2^6$$

$$= 2^{11} - 2^6$$

$$= 2048 - 64 = 1984$$

[NAT]

$$|A| = 12$$



#Q. Let $A = \{1, 2, 3, 4, 5, 7, 8, 10, 11, 14, 17, 18\}$. How many subsets of A contain six elements? 924 Ans ✓

$$\# \text{ Subsets of size } = 6 = {}^n C_6 = {}^{12} C_6 = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$${}^n C_r = \frac{n!}{(n-r)! r!}$$

$$= 11 \times 2 \times 3 \times 2 \times 7$$

$$= 77 \times 12$$

$$= 924$$

[NAT]

$$|A| = 12$$

in set A, no. of even elements = 6
no. of odd elements = 6



#Q. Let $A = \{2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15\}$

How many six-elements subsets of A contain four even integers and two odd integers?

225

Ans

We need to create
subsets of size = 6
with 4 even elements
2 odd elements

= Choose 4 even
elements out of 6 even elements

Choose two
odd elements
out of 6
odd elements

$$= {}^6C_4 * {}^6C_2$$
$$= 15 * 15 = 225$$

[NAT]



#Q. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. How many subsets of A contain only odd integers? 63 does not contain any odd integer
Ans $\therefore$ it must not be included

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$
$$2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1$$

$$= 2^6 = 64 - 1 = 63$$

also include $\{\}$ Exclude $\{\}$

[MCQ]



#Q. Suppose A, B, C and D are subsets of U (the universe) with A as a subset of B and C as subset of D i.e $A \subseteq B$ and $C \subseteq D$, then consider the following statements

I. $A \cap C \subseteq B \cap D$

II. $A \cup C \subseteq B \cup D$

Which of the following is correct options?

A

Only I is true

B

Only II is true

C

Neither I nor II is true

D

Both I and II are true



$$a \in A \cap C$$

i.e. $a \in A$ and $a \in C$

We know $A \subseteq B \therefore a \in B$

& $C \subseteq D \therefore a \in D$

Hence

$$a \in B \cap D$$

[NAT]



even elements = 7
odd elements = 8

#Q. Let $A = \{1, 2, 3, \dots, 15\}$. How many subsets of A contains all of the odd integers in A ? = 128 Ans

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 $1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1$

$$= 2^7 = 128$$

$\therefore$ # Subsets that contain all odd integers = $\underbrace{\binom{8}{8}}_{\text{Choose all 8 out of 8}} \times \underbrace{\left(\binom{7}{0} + \binom{7}{1} + \binom{7}{2} + \dots + \binom{7}{7} \right)}_{\text{Possible Choices for even elements}}$
 $= 1 \times 2^7 = 128$

[MCQ]

are two sets



#Q. Let A, B where $A = \{x \mid x^2 - 7x = -12\}$ and $B = \{x \mid x^2 - x = 6\}$. Determine $A \cup B$ and $A \cap B$.

$$x^2 - 7x + 12 = 0$$

$$x = 3 \text{ \& } x = 4$$

$$A = \{3, 4\}$$

$$x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$x = 3 \text{ \& } x = -2$$

$$B = \{3, -2\}$$

A $A \cup B = \{5\}$ and $A \cap B = \{-2, 3, 4\}$

B $A \cup B = \{3\}$ and $A \cap B = \{-2, 3, 4\}$

C $A \cup B = \{-2, 3, 4\}$ and $A \cap B = \{3\}$

D $A \cup B = \{2, 3, 4\}$ and $A \cap B = \{5\}$

$$\begin{cases} A \cup B = \{3, 4, -2\} \\ A \cap B = \{3\} \end{cases}$$



THANK - YOU